

Supplementary Material for: A Reproducible Software Workflow for Compact Pauli Analysis of Molecular Vibrational Hamiltonians

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I. ADDITIONAL REPRESENTATIVE SPECTRA

II. STAGE-RESOLVED PATULIN BENCHMARK

III. SEPARATE THREE-MOLECULE END-TO-END TIMING CONTROL

A prior integrated validation script (different from the final Pauli-compression code) produced the following timing controls. They are retained only to prevent unsupported speedup claims and should not be merged numerically with the final-code benchmark without rerunning all molecules under the same implementation and hardware conditions.

IV. PROTEIN SCALING STRESS TEST

A separate 1CRN protein benchmark contained 642 atoms and 1926 Cartesian degrees of freedom. A matrix-free reduced Krylov calculation used a primary 64-

dimensional subspace (6 reduced qubits), compared with an 11-qubit full-space index-register count. The run reported a 90.03% reduction in the number of force calls relative to the finite-difference full-Hessian estimate, 28.30 MiB of full Hessian storage versus 0.0312 MiB for the reduced operator, and a same-reduced-operator ST4 $r = 2$ spectral cosine of 0.99999998. However, Dense-NMA versus the primary Lanczos spectrum had cosine 0.8643. This result is therefore a scalability stress test, not evidence that the current 64-dimensional reduction replaces full protein NMA. Rank convergence against the dense reference is required before such a claim.

V. FILES INCLUDED WITH THIS PROJECT

The archive includes the current benchmark code snapshot, the ten-molecule manifest, derived resource table, Patulin Pauli-compression CSV, Patulin Trotter-convergence CSV, and Patulin stage-resolved timing CSV. These machine-readable files should be deposited in a public repository and assigned a DOI before submission.

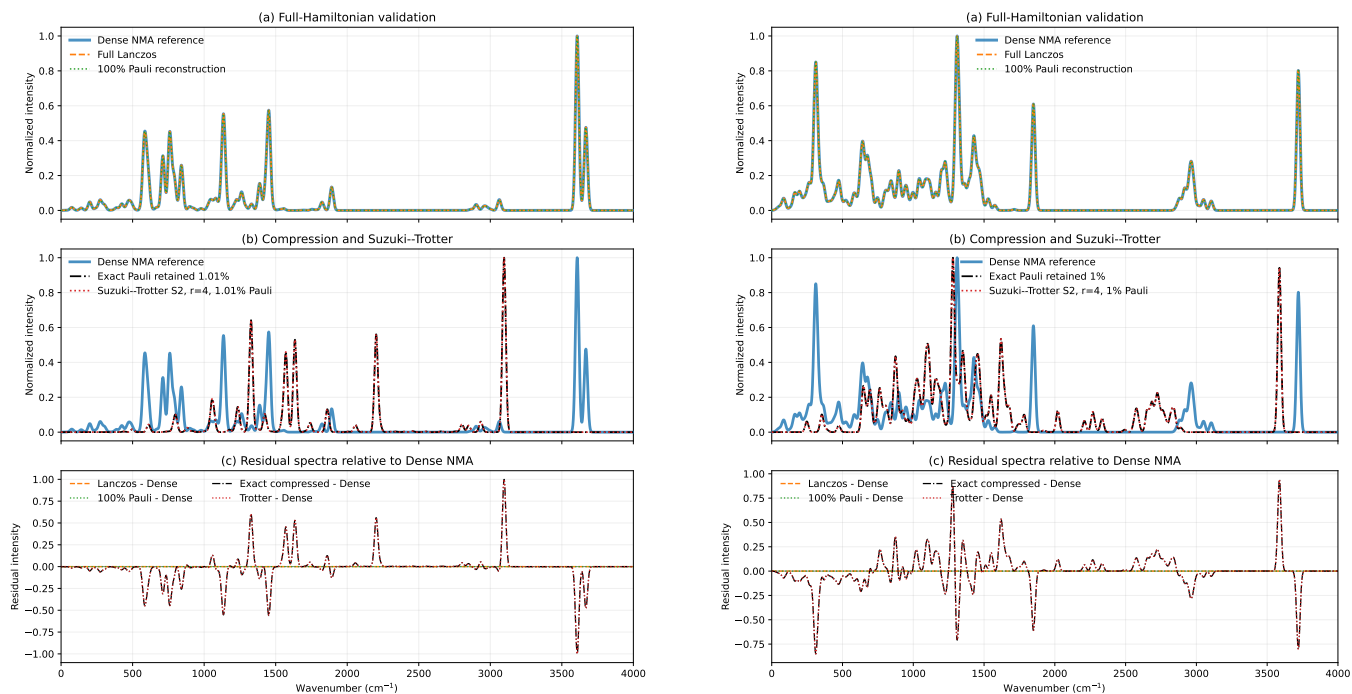


FIG. 1. Citrinin (left) and T-2 toxin (right) full-Hamiltonian, 1% compression/Trotter, and residual panels.

TABLE I. Measured Patulin wall time and RSS changes from the current final-code smoke test. RSS delta is a process-level measurement, not a peak-allocation profiler.

Stage	Time (s)	RSS before (MiB)	RSS after (MiB)
3D preparation/optimization	0.02084	199.55	207.95
Finite-difference Hessian	0.01862	208.41	208.41
Dense NMA	0.000576	208.41	209.43
Full Lanczos	0.001188	210.00	210.46
Low-mode eigsh	0.001823	210.46	210.96
Normalize/pad	0.000163	211.43	211.43
Pauli decomposition	0.03994	211.43	211.73
100% Pauli reconstruction	0.03336	211.73	211.98
10% Pauli reconstruction	0.003476	213.36	213.45
1% Pauli reconstruction	0.000504	213.50	213.58
Exact exp., 1% same H	0.000188	236.80	236.95
ST2, 1%, $r = 1$	0.001352	237.61	237.61
ST2, 1%, $r = 2$	0.002550	238.03	238.03

TABLE II. Historical timing control from the uploaded run log. The reported speedup metric is below unity in all cases; therefore the current architecture does not demonstrate end-to-end speedup over classical NMA.

Molecule	NMA eig. (s)	Krylov (s)	ST4 $r = 2$ (s)	Speedup metric
Anatoxin-a	0.01119	1.7069	0.7408	0.2515
Aflatoxin B1	0.00546	1.5135	0.9197	0.3523
Curcumin	0.00813	1.9554	2.4023	0.3245